

U.S. Electrical Equipment

Breaking the Data Center Labor Ceiling: How Modular Unlocks Capacity, Rewards Vertically Integrated OEMs & Contractors



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Craft labor scarcity is forcing the data center industry to reinvent how it builds. There are 2 layers to this story: 1) the industry is moving from traditional stick-built to modular construction, which shifts work from the job site to the factory, cuts build time and unlocks construction capacity. Second, non-hyperscalers will likely account for a bigger share of future builds, which hands more modular engineering and procurement decisions to the power & cooling OEMs and MEP contractors. Together, these changes reallocate the profit pool towards the most vertically integrated power & cooling OEMs and MEP contractors. Labor is the constraint. Modular is the unlock. Vertical integration captures the value.

Raising the data center construction speed limit by going modular. At the current Mechanical / Electrical / Plumbing (MEP) labor recruitment rates, [US data center builds can increase from 12 GW/year in 2026 to 35 GW/year by 2030](#), which is below the build rate potential of other parts of the supply chain (power generation, GPUs, etc.) To unlock throughput, the industry is shifting from the traditional “stick built” construction approach to modular. Power and cooling system construction is where modular is gaining ground.

5 reasons why modular adoption grows: **1)** increases time to market; **2)** lowers execution risk; **3)** increases predictability; **4)** the economics become more attractive as data center rack density increases. **5)** the transition from 54V AC to 800V DC architecture will cut MEP field labor productivity. The mix of modular vs. stick built is set to grow from 20/80 in 2020 to 60/40 by the end of the decade.

As non-hyperscalers grow as a share of future builds, engineering and procurement decisions will shift towards MEP contractors and OEMs. Non-hyperscalers will likely grow from 45% to 80% of builds by the end of the decade. Compared to hyperscalers, this customer base tends to be less sophisticated and smaller scale, which results in a different procurement approach; they are more likely to outsource the engineering and cede the equipment decisions to the contractors or OEMs. Engineering becomes an additional service to sell for MEP contractors and the content per MW for OEMs naturally grows as equipment procurement decisions move closer to them.

These 2 changes put the most vertically integrated MEP contractors and OEMs at a massive competitive advantage. It enhances their economics: **1)** Faster growth - more data centers will get built; **2)** Bigger wallet share - MEP contractors will increasingly sell engineering, fabrication and installation as a turnkey solution, while OEMs will outfit power and cooling modules with their own equipment as procurement decisions become more systems than product-based; **3)** Higher margins - the value of labor cost/time savings transfer from the field to the factory and engineering department, compensation for higher execution risk; **4)** Increases barrier to entry - shift from product- to systems-based procurement makes it difficult for limited product line companies to compete.

The best positioned: **ETN, SU** and **VRT** are the most vertically integrated OEMs: electrical, thermal and DSIM. **LGN** and **PWR** are the most vertically integrated contractors and can handle an entire MEP project.

BERNSTEIN TICKER TABLE

			12 Aug	TTM		Adjusted EPS				Adjusted P/E (x)		
			2026									
			Closing	Price	Rel.							
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
ETN (Eaton)	O	USD	459.96	534.00	6.4%	USD	12.07	13.45	16.37	38.1	34.2	28.1
HUBB (Hubbell)	O	USD	512.90	584.00	(1.5)%	USD	18.33	20.50	24.03	28.0	25.0	21.3
SU.FP (Schneider)	O	EUR	311.00	350.00	19.8%	EUR	8.43	10.42	13.24	36.9	29.8	23.5
SIE.GR (Siemens)	O	EUR	281.60	330.00	1.3%	EUR	12.23	11.29	11.64	28.2	24.3	22.9
ENR.GR (Siemens Energy)	O	EUR	161.72	210.00	35.7%	EUR	1.63	4.24	5.94	39.2	21.8	15.6
ABBN.SW (ABB)	M	CHF	83.48	80.00	34.1%	USD	2.56	3.02	3.63	40.1	34.0	28.3
LR.FP (Legrand)	O	EUR	142.70	170.00	(11.5)%	EUR	5.22	6.13	6.86	27.3	23.3	20.8
VRT (Vertiv)	O	USD	288.36	368.00	80.4%	USD	4.20	6.60	9.06	68.7	43.7	31.8
NVT (nVent)	O	USD	171.16	220.00	66.2%	USD	3.34	4.84	6.26	51.2	35.4	27.3
TT (Trane)	O	USD	478.55	555.00	(11.2)%	USD	13.06	14.98	17.82	36.6	31.9	26.9
JCI (Johnson Controls)	O	USD	152.79	173.00	23.1%	USD	3.78	4.97	5.93	40.4	30.7	25.7
CARR (Carrier)	M	USD	63.08	78.00	(24.2)%	USD	2.57	2.85	3.29	24.5	22.1	19.2
IREN (IREN)	O	USD	43.67	100.00	124.7%	USD	0.41	(0.14)	(1.04)	106.5	(304.6)	(41.8)
WULF (TeraWulf)	O	USD	17.19	36.00	207.8%	USD	(1.66)	(1.66)	(0.71)	(10.3)	(10.4)	(24.1)
CORZ (Core Scientific)	O	USD	20.87	32.00	17.9%	USD	(0.88)	(0.92)	0.26	(23.7)	(22.6)	80.3
CIFR (Cipher)	O	USD	17.87	32.00	258.4%	USD	(2.15)	(1.13)	(0.53)	(8.3)	(15.8)	(33.7)
RIOT (Riot)	O	USD	20.32	35.00	57.4%	USD	(1.95)	0.21	(1.24)	(10.4)	94.8	(16.4)
CLSK (Cleanspark)	O	USD	12.18	24.00	2.6%	USD	1.25	0.08	1.07	9.7	145.7	11.4
MARA (MARA)	M	USD	9.64	17.00	(58.9)%	USD	(3.69)	3.29	(1.22)	(2.6)	2.9	(7.9)
DLR (Digital Realty)	O	USD	197.06	226.00	(3.8)%	USD	3.87	3.09	3.21	50.9	63.7	61.4
EQIX (Equinix)	O	USD	1,069.17	1,270.00	15.6%	USD	14.96	17.79	20.87	71.5	60.1	51.2
CSQR (Csquare)	O	USD	22.98	27.00	NA	USD	0.00	(1.34)	(1.06)		(17.1)	(21.6)
LGN (Legence)	O	USD	68.68	103.00	NA	USD	298.82	479.93	567.27	29.4	18.3	15.5
PWR (Quanta Services)	M	USD	678.02	748.00	52.9%	USD	10.75	16.90	19.24	63.1	40.1	35.2
SPX			7,748.50									
EDME			1,636.48									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

SIE.GR, ENR.GR, IREN, WULF, CORZ, CIFR, RIOT, CLSK, MARA estimate is Reported EPS; LGN estimate is EBITDA (M); ENR.GR, LGN valuation is EV/EBITDA (x); IREN, WULF, CORZ, CIFR, RIOT, CLSK, MARA valuation is Reported P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

US Machinery and Electricals: We maintain our Outperform rating for LGN, ETN and HUBB and their respective PT of \$103, \$534 and \$584. We maintain Market-Perform rating of PWR and its PT of \$748.

European Capital Goods: We rate Schneider (PT €350), Siemens (€330), Legrand (€170), and Siemens Energy (€210) Outperform, and rate ABB (CHF 80) Market-Perform.

US Multi-Industrials: We rate VRT (PT \$368), NVT (PT \$220), TT (PT \$555) and JCI (PT \$173) Outperform. We rate CARR (PT \$78) Market-Perform.

US Communications Infrastructure: We rate DLR, EQIX, and CSQR Outperform.

Emerging AI Infra/Bitcoin Miners: Crypto miners remain well positioned to solve 'time to compute' given their planned 30+ GW power portfolio and operating ability to deliver 'warm powered shells' in time. Over the past two years, miners have contracted over 8 GW of power capacity to hyperscalers, neoclouds, leading AI labs and AI-chip manufacturers across deals worth over \$170Bn. We continue to like IREN as a vertically integrated neocloud. We rate IREN Outperform (PT \$100). We rate WULF Outperform (PT\$36), CIFR Outperform (PT\$32), CORZ Outperform (PT\$32), CLSK Outperform (PT \$24), RIOT Outperform (PT \$35) and MARA Market-Perform (PT \$17).

DETAILS

Craft labor scarcity is forcing the data center industry to reinvent how it builds, and there are 2 layers to this story. First, the industry is moving from traditional stick-built to modular construction, which shifts work from the job site to the factory, cuts build time and unlocks construction capacity. Second, non-hyperscalers will likely account for a bigger share of future builds, which hands more engineering and procurement decisions to the power & cooling OEMs and MEP contractors. Together, these changes reallocate the profit pool towards the most vertically integrated power & cooling OEMs and MEP contractors who can design, fabricate and install the entire system. Labor is the constraint. Modular is the unlock. Vertical integration captures the value.

Craft labor scarcity is changing how data centers get built. At the current Mechanical / Electrical / Plumbing (MEP) labor recruitment rates, [US data center builds can increase from 12 GW/year in 2026 to 35 GW/year by 2030](#), which is below the build rate potential of other parts of the supply chain (power generation, GPUs, etc.) To unlock greater construction throughput, the industry is shifting from the traditional “stick built” construction approach where all materials and components arrive at the job-site and the craft labor builds everything onsite, to pre-fabricated/modular approach, where the craft labor builds modules at the factory, ships them to the job site, where a far less complex final assembly occurs. Power and cooling system construction is where modularization is gaining the most ground.

5 reasons why modular construction will grow: **1)** increases speed to market – factory production can run in parallel with field construction and the build timeline can fall by as much as 60%; **2)** lowers execution risk – the most complex work moves to the factory, a controlled workspace; **3)** increased predictability – standard designs, factory testing cut mistakes.; **4)** the economics of modular become more attractive as data center rack density increases. **5)** the transition from 54V AC to 800V DC architecture will cut MEP field labor productivity - check with an MEP estimator suggest an initial 50% increase in MEP labor hours in the fit out stage to deal with the greater complexity, safety challenges, etc. As such, the mix of modular vs. stick built is growing from 20/80 in 2020 to 40/60 today and possibly to 60/40 by the end of the decade.

As non-hyperscalers account for a greater share of the data center build, engineering and procurement decisions will shift towards MEP contractors and the power & cooling OEMs. Going forward, hyperscalers will likely represent a shrinking percentage of the build, falling from 55% of the construction activity in 2024, to 40% today and may fall to ~20% in a few years if the current project pipeline mix holds. This change matters because hyperscalers have the most sophisticated modular procurement process, with in-house engineers and full control of the equipment selection, where they are comparatively more focused on best of breed rather than best of suite. In contrast, the other data center operator types (colocators, developers, neoclouds, crypto) may be less sophisticated (more reliant on 3rd party engineering, less buying power, greater willingness cede equipment purchasing decisions to the contractor or even the power & cooling OEM(s). From these customers' perspectives, there is tremendous value in engaging early with engineering & design and even using the same firm to fabricate and install the modules. There is also value in narrowing the vendor list for procuring the individual equipment that make up the modules. All of this lowers project execution risk, raises schedule certainty and drives repeatability for the customer.

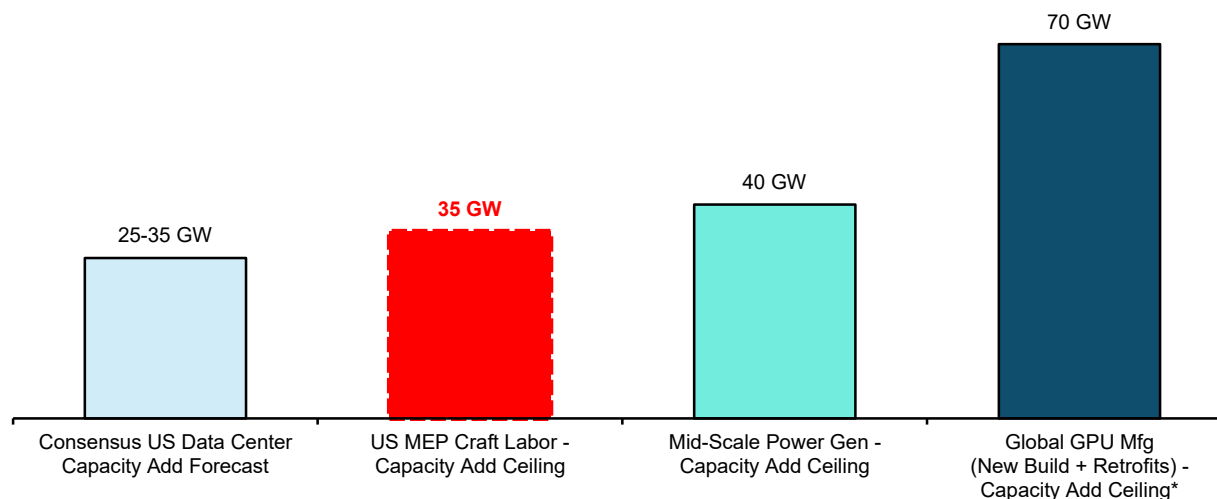
The most vertically integrated MEP and power & cooling OEMs are at a massive competitive advantage from this customer mix shift, given they can serve as the one-stop shop for all equipment and software needs. Moreover, the move towards modularization enhances MEP contractor and OEM economics: **1)** More data centers will get built because modularization expands throughput expands at the greatest construction chokepoint; **2)** Wallet share increases for MEP contractors since engineering takes on a bigger role and for electrical and thermal OEMs as customers increasingly procure at the system-level, rather than product; **3)** margins expand for MEP contractors and OEMs to account for the value created by the construction and craft labor time savings that come from modularization, the greater share of execution risk that comes with this shift, and for MEPs contractors only, a higher revenue share of structurally higher engineering work; **4)** this emerging one-stop shop model helps fend off single product new entrants into the power & cooling space. Conversely, the field-labor only contractors have the most to lose from this change.

The best positioned companies: ETN, SU and VRT stand to gain the most in a world that is more modular, given they can cover the 3 most important elements of the data technology stack: electrical, thermal and data center infrastructure management (software overlay). On the MEP contractor side, **LGN** and **PWR** are best positioned given they are the most vertically integrated and can handle an entire MEP project from concept to commissioning (engineering + fabrication + installation).

CRAFT LABOR SCARCITY IS A KEY LIMITING FACTOR OF THE DATA CENTER BUILD OUT

EXHIBIT 1: If MEP Craft Labor Enables 35 GW of US Data Center Builds by 2030, Then It is Enough to Support The Consensus View of 25-35 GW of Annual Capacity Adds, but Just Shy of Supporting The Mid-Scale Power Gen Capacity Build-Out

By 2030

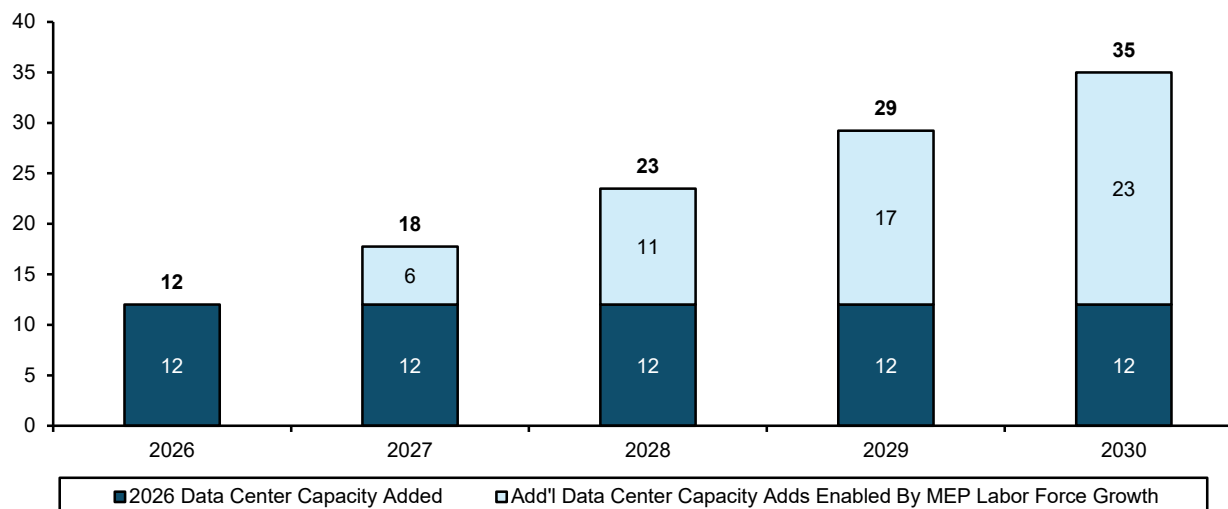


*Courtesy of Bernstein Japan & EU Semiconductor Team. Research Note Link: Infineon/Renesas: CPU renaissance emerges as a second engine for power semi demand, set to outstrip supply

Source: Gartner, Bloomberg, Bernstein estimates and analysis

EXHIBIT 2: At Current MEP Craft Labor Recruitment Rates, US Data Center Builds Could Grow by 6 GW/year or at a ~30% CAGR from 12 GW in 2026 to 35 GW by 2030

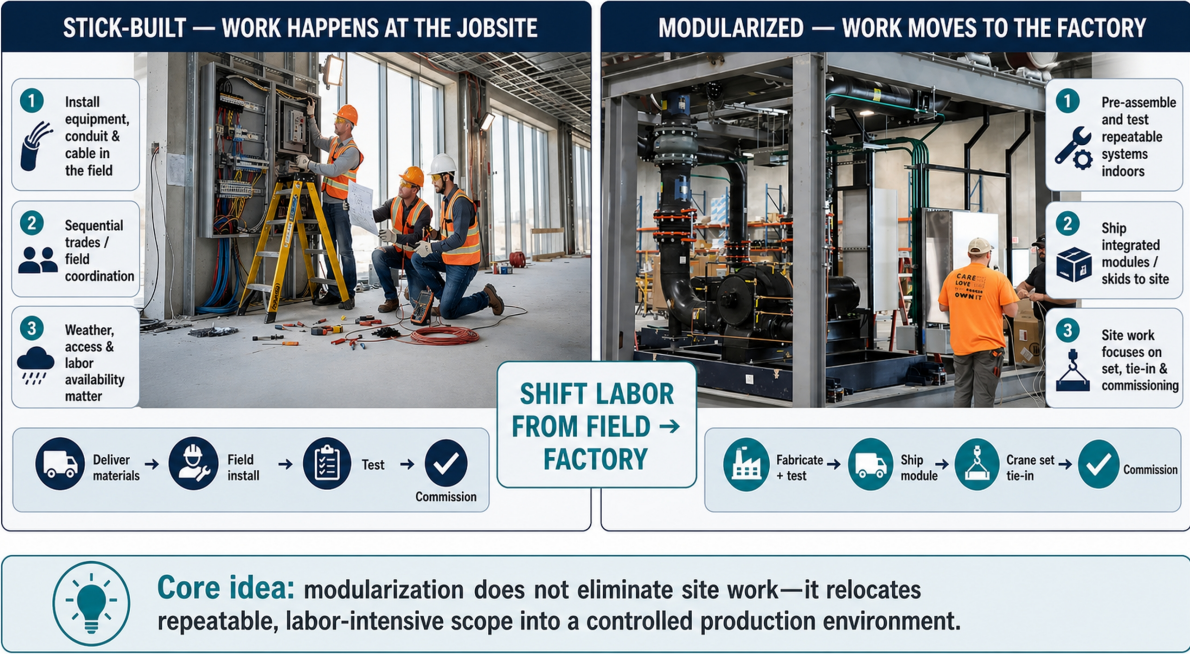
- Data Center Capacity Add Ceiling -
As Determined by MEP Craft Labor Recruitment (GW)



Source: Aterio, Company Reports, Bernstein estimates and analysis

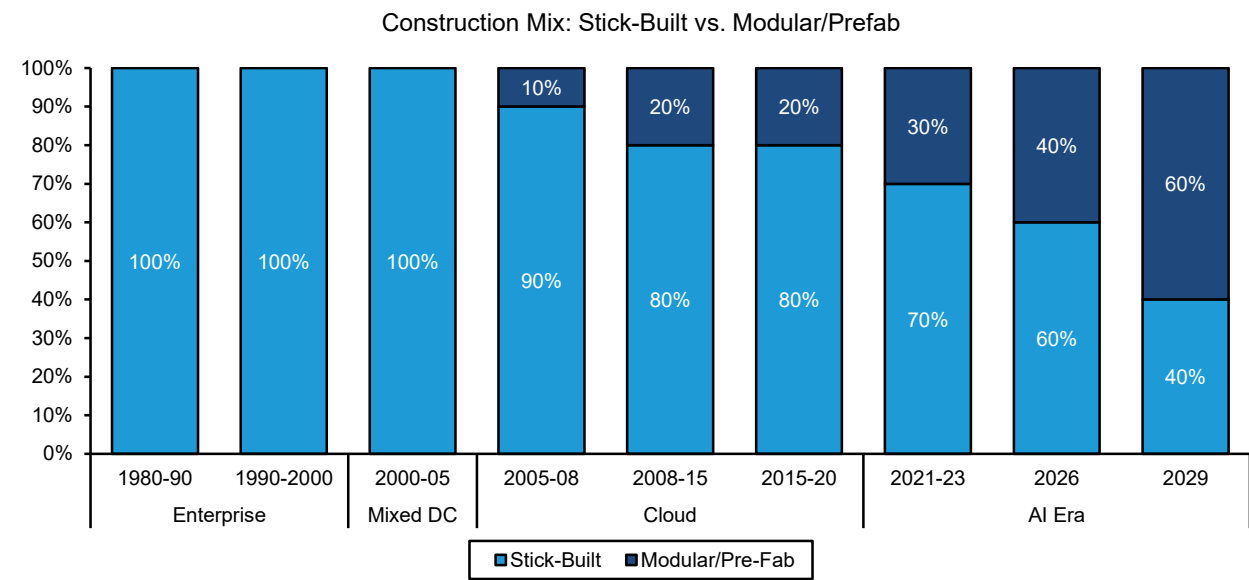
WHAT IS MODULAR CONSTRUCTION?

EXHIBIT 3: Modularization Enables a Lego-Style Approach to Data Center Construction



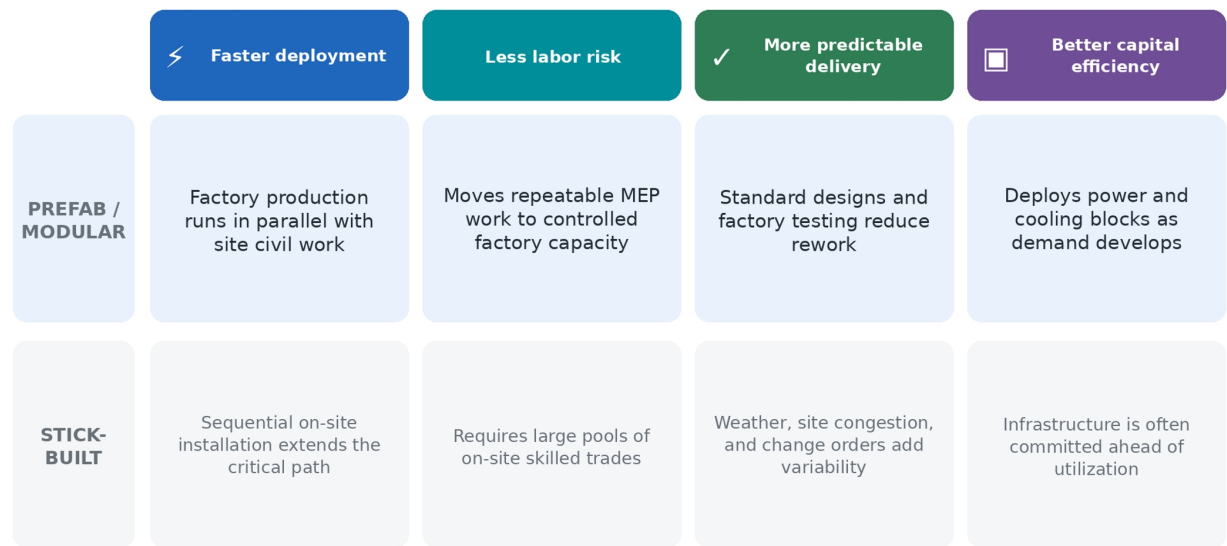
UNLOCKS THROUGHPUT AND BECOMING THE PREFERRED WAY TO BUILD DATA CENTERS

EXHIBIT 5: **Modular Is Becoming The Dominant Data Center Construction Approach: From 20% to 60% of All Builds In This Decade**



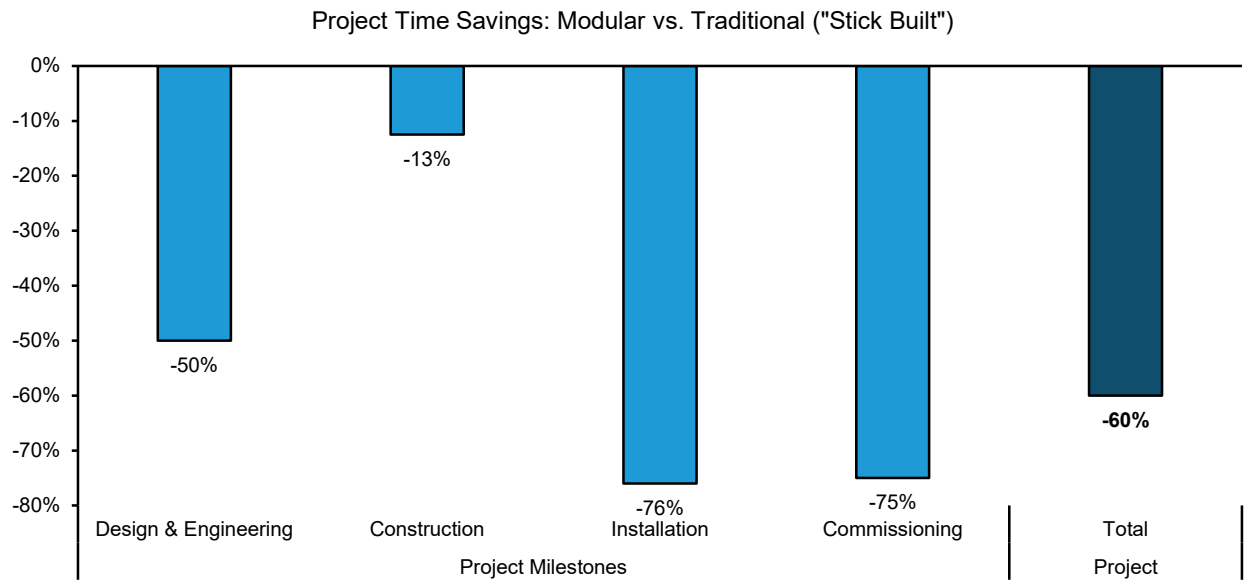
Source: Rosendin, Bernstein analysis

EXHIBIT 6: **4 Benefits of Transitioning from the Traditional Stick-Built to Modular Construction Approach**



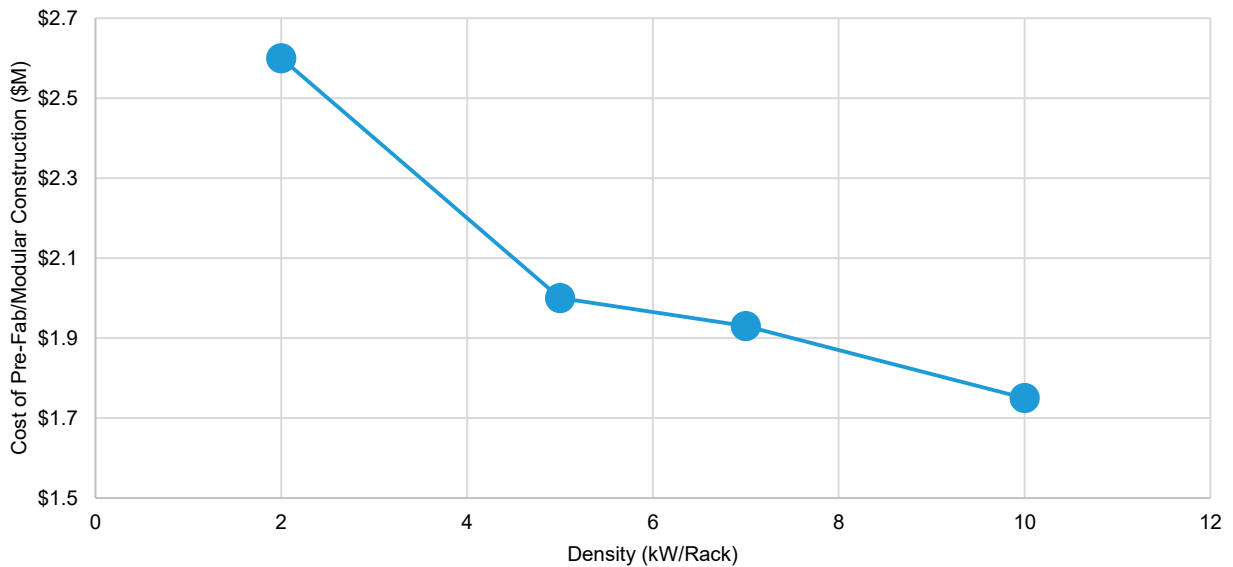
Source: Bernstein analysis

EXHIBIT 7: Above All Else, The Time Savings from Going Modular Can Be Considerable: Data Centers Can Be Built By As Much As 60% Faster



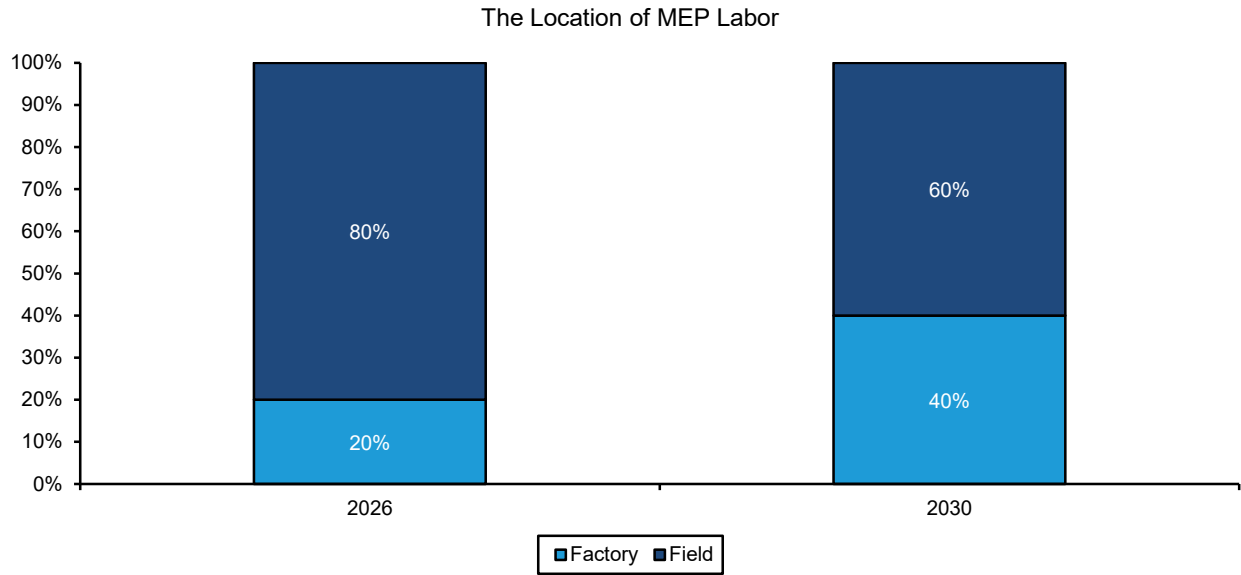
Source: Schneider Electric, Bernstein analysis

EXHIBIT 8: More Adoption Ahead: As Rack Density Rises, The Cost of Modular Construction Falls



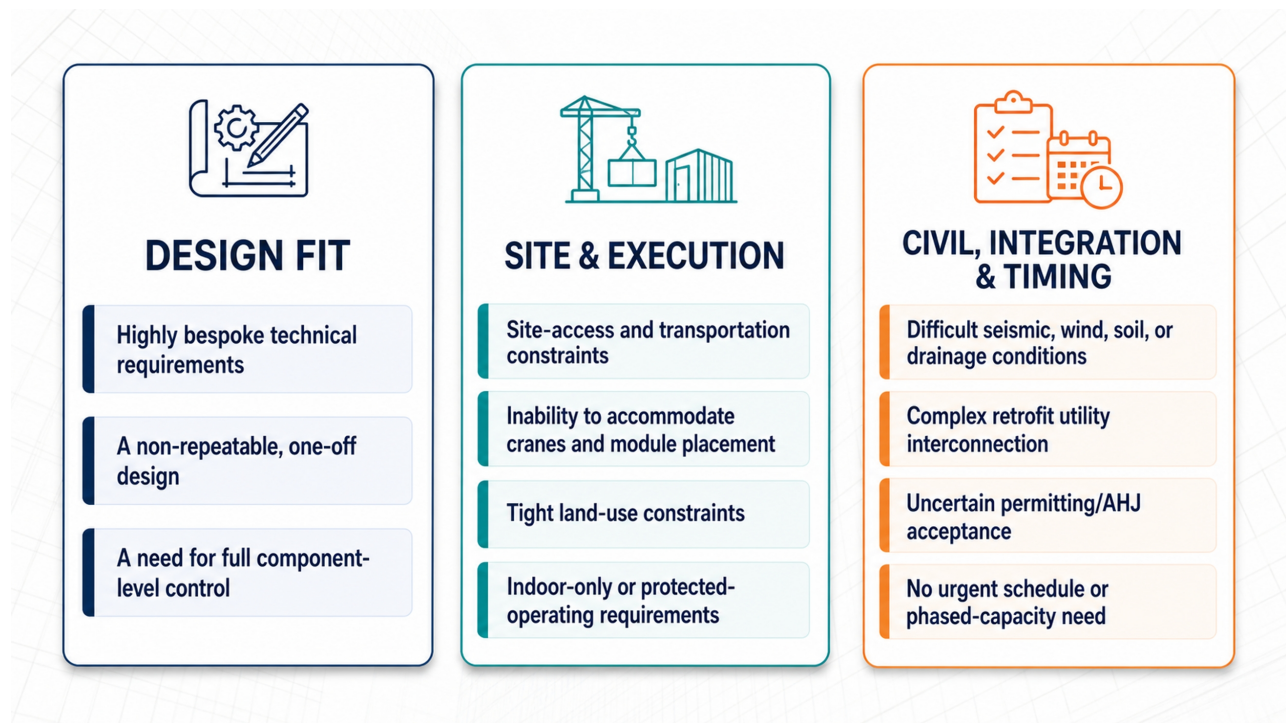
Source: Schneider Electric, Bernstein analysis

EXHIBIT 9: **Greater Modularization Means that MEP Craft Labor Will Migrate From the Field To Factory**



Source: Rosendin, Bernstein analysis

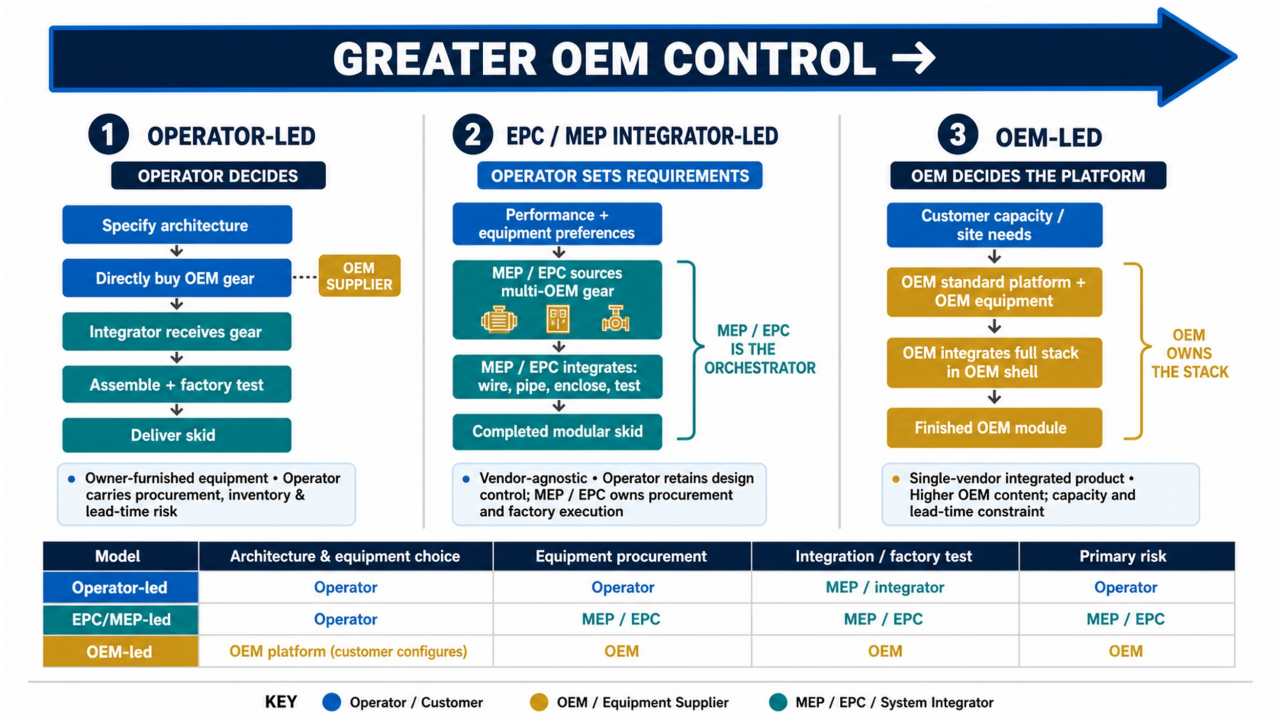
EXHIBIT 10: **Reasons To Remain With Traditional Stick-Built Instead of Modular Constructions**



Source: Schneider Electric, Bernstein analysis

AS MORE NON-HYPERSCALERS DRIVE THE BUILD, THE MODULAR ENGINEERING AND EQUIPMENT PROCUREMENT DECISIONS WILL SHIFT TOWARDS MEP CONTRACTORS AND OEMS

EXHIBIT 11: Modular Construction Projects Take 3 Approaches to Equipment Procurement Decision: The More OEM-Led, The More Vertically Integrated the Module, Which Maximizes the TAM



Source: Semi Analysis, Bernstein analysis

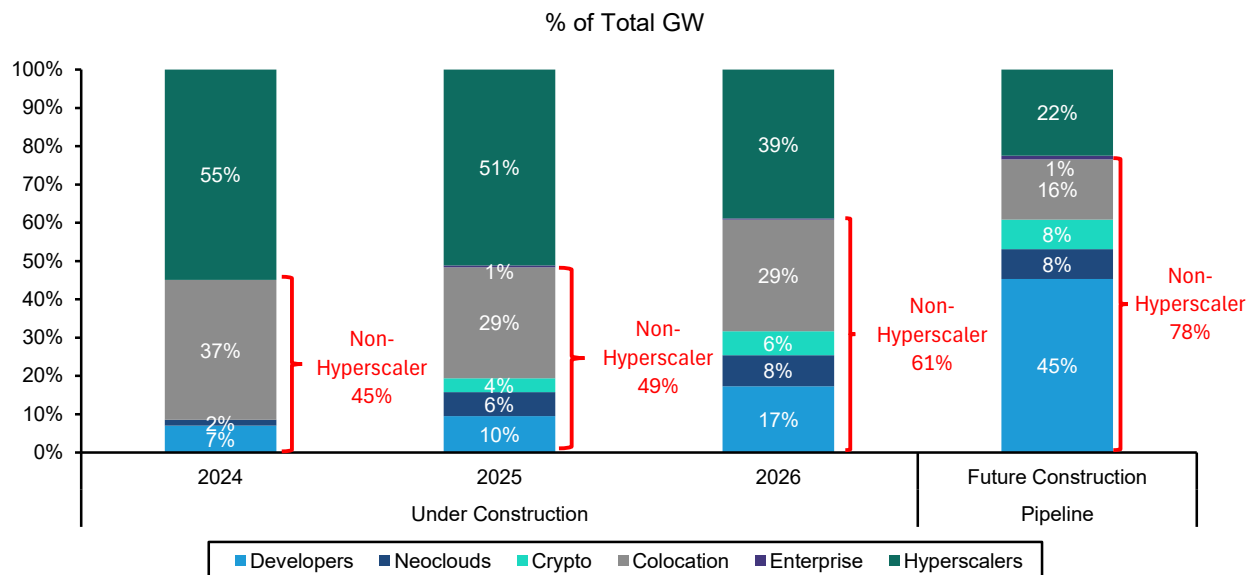
EXHIBIT 12: **Operator Preferences on Modular Procurement Approach (Directional)**

Customer type	Operator-led	EPC / MEP integrator-led	OEM-led platform
Hyperscaler	● ● ● ● ●	● ● ● ● ●	● ● ○ ○ ○
Colocator	● ● ● ○ ○	● ● ● ● ●	● ● ● ● ○
Developer	● ● ○ ○ ○	● ● ● ● ●	● ● ● ○ ○
Neocloud	● ● ○ ○ ○	● ● ● ● ○	● ● ● ● ●
Crypto	● ○ ○ ○ ○	● ● ● ○ ○	● ● ● ● ●

Relative frequency ● ● ● ● ● Very frequent ● ● ● ● ○ Frequent ● ● ● ○ ○ Moderate ● ● ○ ○ ○ Less frequent ● ○ ○ ○ ○ Least frequent

Sophisticated customers preserve design authority and use MEP / EPCs for execution; speed- and standardization-oriented customers more often purchase OEM-led platforms.

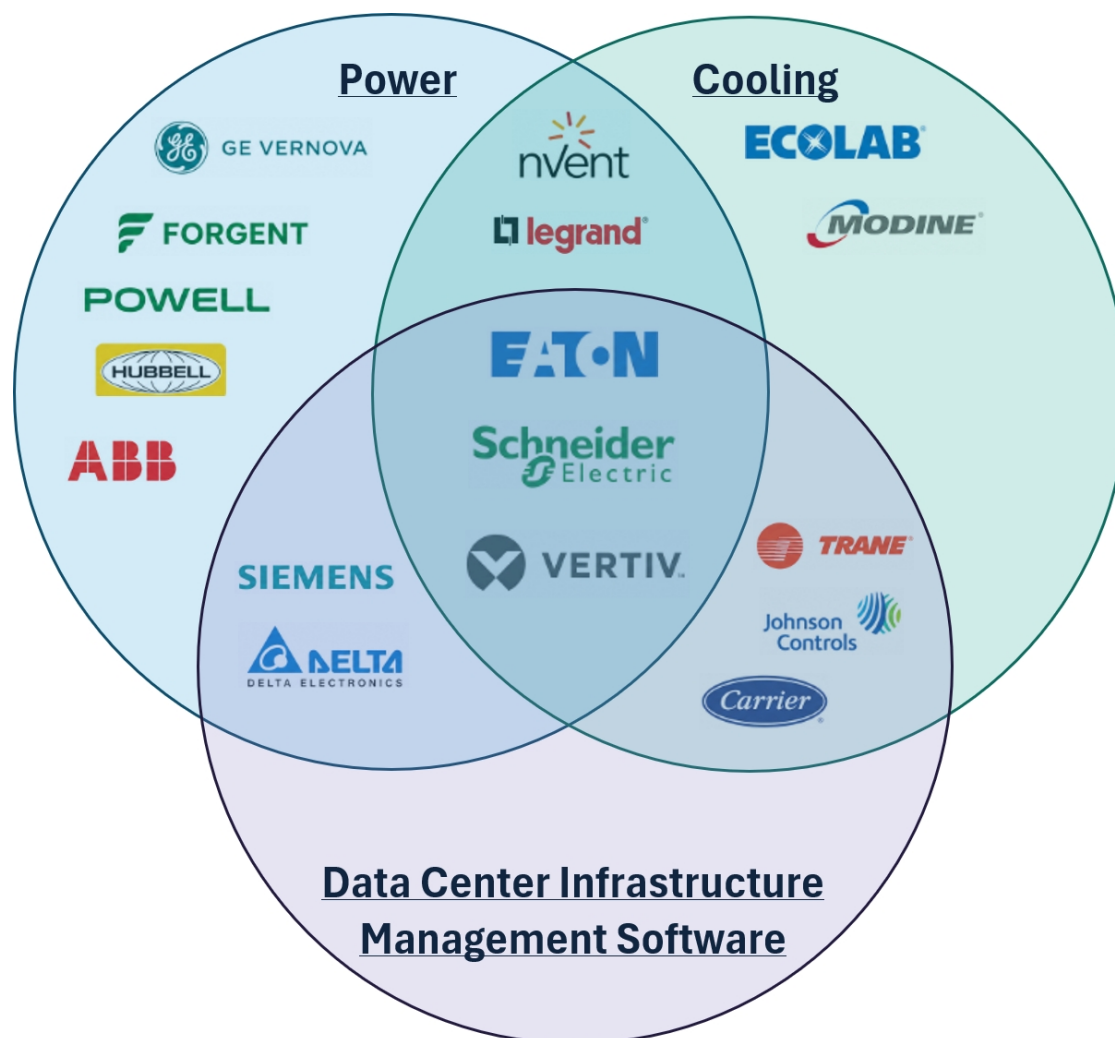
Source: Semi Analysis, Bernstein analysis

EXHIBIT 13: **As Non-Hyperscalers Grow As a Share of Data Center Build, The Modular Procurement Model Will Become More OEM & MEP Led**

Source: Aterio, Bernstein analysis

THE MOST VERTICALLY INTEGRATED OEMS AND MEP CONTRACTORS WILL BENEFIT THE MOST

EXHIBIT 14: **As Equipment Decisions Migrate From Data Center Operators to OEMs, the Most Vertically Integrated Will Capture a Greater Share of the Profit Pool: Eaton, Schneider, Vertiv**



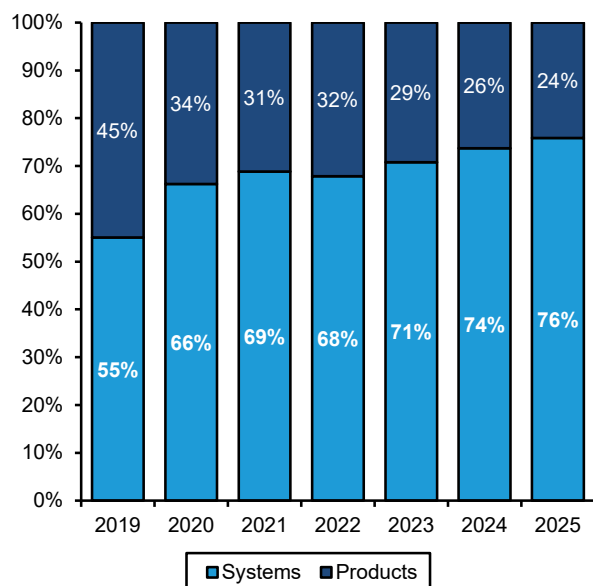
Source: Bernstein analysis

EXHIBIT 15: **As Procurement Shifts From Operator-Led, The Engineering-Modularization Nexus Becomes More Important: MEP Contractors That Offer Turnkey Solutions, Begin The Design Conversation At the Earliest Stages of that Data Center Project Will Increase Wallet Share**

	Legence	Quanta Services	EMCOR	Comfort Systems
Service				
Engineering & Design	✓	✓	✓	
Prefabrication/Modular	✓	✓	✓	✓
Field Construction	✓	✓	✓	✓
Maintain	✓	✓	✓	✓
Craft				
Mechanical	✓	✓	✓	✓
Electrical	✓	✓	✓	✓
Plumbing	✓	✓	✓	✓

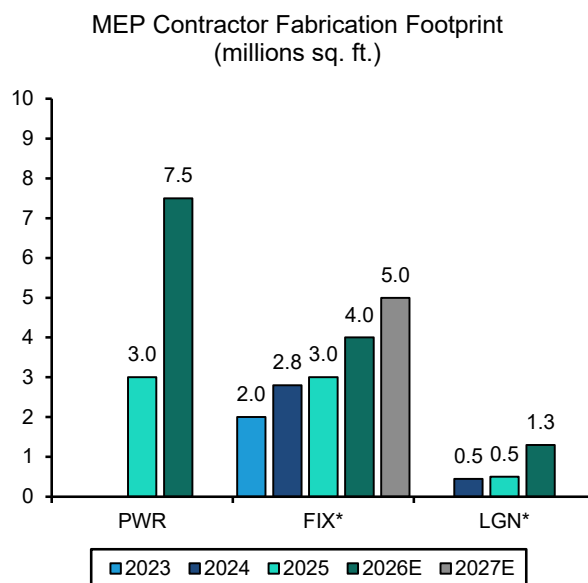
Source: Company data, Bernstein analysis

EXHIBIT 16: **Modularization is Shifting ETN's Electrical Americas Revenue Mix From Single Products To Systems**



Source: Company data, Bernstein analysis

EXHIBIT 17: **Modularization is Pushing MEP Contractors To More Than Double Their Fabrication Footprints**



Source: Company data, Bernstein analysis

EXHIBIT 18: **The Strategic Impact of Modularization on Power & Cooling OEMs and MEP Contractors**

Expands The Data Center TAM	Fewer labor hours enables faster throughput and more data center construction than otherwise achievable via stick-build approach
Increases the Wallet Share At The Edges of the Value Chain	OEMs: more modularization, especailly OEM-led, means procurement decisions will increasingly be made from the perspective of "best of suite" rather than "best of breed", favoring the large vertically integrated OEMs MEPs: Engineering becomes more important as modularization requires early engagement Field Construction Labor: shifting work to onsite manufacturing will cut the number of labor hours
Expands Margins	OEMs: unlocks more value-based pricing opportunities (labor cost savings) and the shift in execution risk towards the OEM will warrant a higher price MEPs: higher mix of engineering work, execution risk shifts their way, value based pricing opportunities (labor cost savings)

Source: Bernstein analysis

EXHIBIT 19: **OEM Modular Power & Cooling Offerings**

	Modular product suite	System	Product Description
Eaton	Modular Data Center / Modular Power Assemblies	Power & Cooling	Engineered outdoor or indoor data-center modules that combine switchgear, UPS and batteries, electrical/mechanical distribution, IT and HVAC. Eaton can configure N, N+1, or 2N power and cooling architectures within a factory-integrated enclosure.
	Fibrebond	Power	Designs and builds engineered-to-order, pre-integrated modular power enclosures for data centers.
	NordicEPOD	Power & Cooling	Standardized, factory-built electrical power modules installed between the grid and critical IT load. EPODs package UPS, switchgear, power distribution, backup systems, cooling, and controls.
	SmartRack Modular Data Centers	Power & Cooling	Smaller rack, cooling, and service enclosures that can be combined into an optimized, maintainable modular IT environment more applicable to edge, enterprise, and contained deployments than campus-scale AI halls. tripp-lite.
Schneider	EcoStruxure Modular Data Center	Power & Cooling	Containerized/prefabricated solution architecture combining power, cooling, and IT infrastructure in pre-engineered, pre-tested modules.
ABB	Prefabricated eHouses and skid-mounted unit substations	Power	Factory-built electrical modules that package transformers, MV/LV switchgear, protection, controls, and ancillary equipment. ABB positions these repeatable power-distribution blocks for enterprise, colocation, cloud, and edge projects.
	PowerBlock	Power & Cooling	A pre-engineered, pre-tested physical-infrastructure system integrating power and cooling equipment; designed to replace a portion of conventional custom-built infrastructure with a faster-deploying standardized block.
Forgent	Power Skids & E-Houses	Power	Factory-assembled, tested platforms combining transformers, switchgear, controls, and supporting distribution equipment.
Siemens	Custom Plug-and-Play Power Pods	Power	Prefabricated and factory-tested power-distribution modules incorporating electrical components, software, and tools in a compact package, intended to shorten energization and reduce construction complexity.
	Modular Edge	Power & Cooling	A partner-delivered, prefabricated modular building solution: Cadolto supplies the modular structure and climate control, while Siemens supplies MV/LV distribution, building automation, fire safety, and security systems.
Legrand	AI Pod	Power & Cooling	An AI-ready row-level infrastructure architecture integrating Starline busway, Raritan in-rack PDUs, and ColdLogik rear-door heat exchangers.
	Girtz Power Pods	Power	Highly customized modular power-integration systems combining electrical and generator infrastructure for backup power or behind-the-meter primary power.
Hubbell	PowerGain Modular Power Units (MPUs)	Power	Modular distribution units that deliver power from remote power panels directly to IT cabinets using direct-to-panel connections; intended to simplify deployment and future reconfiguration in the white space.
	PCX Modular Data Centers	Power	PCX, a Hubbell business, integrates prefabricated electrical and mechanical systems, including modular data centers and indoor/outdoor distribution centers. This is primarily a bespoke power-block and systems-integration offering rather than a standardized container SKU.
Vertiv	SmartMod / SmartMod Max CW	Power & Cooling	Prefabricated all-in-one modular data-center solutions: SmartMod targets lower/medium-density deployments, while SmartMod Max CW is an AI-ready multi-module system supporting high/extreme-density racks using perimeter or rear-door heat-exchanger cooling and direct liquid cooling.
	Vertiv OneCore	Power & Cooling	A hybrid-built, factory-assembled data-center platform integrating power, thermal, and IT infrastructure into repeatable prefabricated modules.
	MegaMod CoolChip / SmartRun	Power & Cooling	MegaMod CoolChip is a high-density AI-oriented modular infrastructure solution; SmartRun is a modular overhead distribution architecture supporting power, liquid cooling, containment, and rack-row buildout at multi-megawatt scale.
nVent	Power Module / PowerNexus	Power	Factory-integrated 480V three-phase modules or skids that package power infrastructure for rapid deployment; Vertiv identifies Power Module configurations up to 30 MW and PowerNexus around 2.5 MW in the cited portfolio.
	Liquid-to-Air (LTA) Sidecar Heat Rejection Unit	Cooling	Modular liquid-to-air CDU/heat-rejection equipment for up to two liquid-cooled racks without requiring a facility-water system. It is targeted at incremental AI retrofits and edge deployments.
JCI	Silent-Aire Modular Data Centers	Cooling	Hyperscale-oriented modular critical-infrastructure offering, combining rapid-deployment modular data-center construction with purpose-built thermal systems.
Trane	Stellar Energy Digital	Cooling	A turnkey modular cooling platform for liquid-cooled data centers. Stellar Energy designs and builds prefabricated modular cooling plants, central utility plants, and coolant distribution units.

Source: Company data, Bernstein analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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